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Fuel cell system

Abstract

When controlling the power generation state of a fuel cell stack based on a generated current command value, a control device sets the generated current command value based on a generated power command value and a generated voltage acquired by a voltage sensor such that the following expression is satisfied, regardless of whether a bleed valve is opened or closed: generated current command value=generated power command value/generated voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-058974 filed on Mar. 31, 2021, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a fuel cell system including a fuel cell that generates power by means of a fuel gas supplied to an anode through a fuel gas supply flow path, and an oxygen-containing gas supplied to a cathode through an oxygen-containing gas supply flow path.

Description of the Related Art

(3) For example, a solid polymer electrolyte fuel cell is provided with a membrane electrode assembly (MEA) in which an anode is disposed on one surface of an electrolyte membrane made of a polymer ion exchange membrane and a cathode is disposed on the other surface of the electrolyte membrane. A power generation cell (unit cell) is formed by sandwiching the membrane electrode assembly between separators.

(4) Usually, a predetermined number of power generation cells are stacked to be incorporated into, for example, a fuel cell vehicle (fuel cell electric vehicle) or the like as a fuel cell stack.

(5) In this type of fuel cell, for efficient use of fuel gas, fuel exhaust gas discharged from the outlet side of the anode of the fuel cell is circulated again into the anode from the inlet side of the anode.

(6) The fuel exhaust gas is mainly formed of unreacted fuel gas and nitrogen gas that permeates through the electrolyte membrane from the cathode side.

(7) In such a configuration, there is a concern about deterioration of the electrolyte membrane, or so-called membrane deterioration, due to a decrease in the concentration of fuel gas (shortage of fuel gas) in the anode.

(8) In order to suppress the occurrence of membrane deterioration, for example, a technique disclosed in JP 2019-114351 A can be used.

(9) In this technique, by opening an on-off valve of a connection flow path, fuel exhaust gas containing nitrogen gas and discharged from the outlet side of the anode of the fuel cell is returned to the inlet side of the cathode via the connection flow path, and circulated to the cathode. In this case, in the cathode, the unreacted fuel gas and oxygen-containing gas supplied to the cathode from the outside react with each other by a catalytic reaction to produce water, and the produced water and the nitrogen gas are discharged to the outside from the outlet side of the cathode.

(10) Thus, it is possible to suppress a decrease in the concentration of the fuel gas flowing through the anode due to an increase of the nitrogen gas on the anode side.

SUMMARY OF THE INVENTION

(11) However, in the technique disclosed in JP 2019-114351 A, when the on-off valve of the connection flow path is opened, the voltage of the power generation cell decreases because of a shortage of the oxygen-containing gas that contributes to the power generation on the cathode side, due to the catalytic reaction.

(12) Therefore, in a fuel cell system in which the generated current of the fuel cell is controlled based on a generated current command value, there is a problem that the generated power decreases.

(13) The present invention has been made in view of such a problem and an object thereof is to provide a fuel cell system capable of avoiding or suppressing a change in generated power even when a generated voltage is lowered by opening an on-off valve of a connection flow path for supplying fuel exhaust gas to the inlet side of a cathode.

(14) According to one aspect of the present invention, provided is a fuel cell system comprising a fuel cell configured to generate power using a fuel gas supplied to an anode through a fuel gas supply flow path and an oxygen-containing gas supplied to a cathode through an oxygen-containing gas supply flow path, the fuel cell system further comprising: a fuel exhaust gas flow path through which a fuel exhaust gas discharged from the anode flows; a connection flow path configured to allow the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other; an on-off valve configured to open or close the connection flow path; a control device configured to control a state of supply of the fuel gas to the fuel cell and a state of supply of the oxygen-containing gas to the fuel cell; and a voltage acquirer configured to acquire a generated voltage of the fuel cell, wherein, when controlling a power generation state of the fuel cell based on a generated current command value, the control device sets the generated current command value based on a generated power command value and the generated voltage acquired by the voltage acquirer.

(15) According to another aspect of the present invention, provided is a fuel cell system comprising a fuel cell configured to generate power using a fuel gas supplied to an anode through a fuel gas supply flow path and an oxygen-containing gas supplied to a cathode through an oxygen-containing gas supply flow path, the fuel cell system further comprising: a fuel exhaust gas flow path through which a fuel exhaust gas discharged from the anode flows; a connection flow path configured to allow the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other; an on-off valve configured to open or close the connection flow path; a control device configured to control a state of supply of the fuel gas to the fuel cell and a state of supply of the oxygen-containing gas to the fuel cell; and a voltage acquirer configured to acquire a generated voltage of the fuel cell, wherein, when controlling a power generation state of the fuel cell based on a generated current command value, the control device performs first power generation control in which the generated current command value is set based on a generated power command value during closing of the on-off valve, and performs second power generation control in which the generated current command value is set based on the generated power command value and the generated voltage during opening of the on-off valve.

(16) According to one aspect of the present invention, even when the on-off valve is opened to cause the fuel exhaust gas to flow through the cathode and the generated voltage of the fuel cell therefore decreases, a change in the generated power can be avoided or suppressed since the generated current command value is increased based on the decrease in the generated voltage. Since the power generation control is not changed in this control, the control can be simplified.

(17) According to the other aspect of the present invention, during power generation, when the concentration of the fuel gas flowing through the anode decreases due to the influence of nitrogen gas permeating from the cathode side to the anode, and the on-off valve of the connection flow path allowing the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other is therefore opened, the generated current command value is set based on the generated power command value and the generated voltage. Therefore, the generated power of the fuel cell does not change or the change of the generated power of the fuel cell can be suppressed even if the fuel gas concentration control (fuel gas concentration decrease prevention control) at the anode is performed.

(18) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic block diagram showing the configuration of a fuel cell vehicle in which a fuel cell system according to a first embodiment is incorporated;
- (2) FIG. 2A is a schematic block diagram of a current converter, which is one function of a control device shown in FIG. 1 according to the first embodiment; and FIG. 2B is a schematic block diagram of a current converter, which is another function of the control device in FIG. 1 according to a second embodiment;
- (3) FIG. 3 is a block diagram showing, by arrows, a current flow and a gas flow in the fuel cell system of FIG. 1;
- (4) FIG. 4 is a flowchart for explaining the operation in the first embodiment;
- (5) FIG. 5 is a time chart showing a correspondence relationship between opening and closing of a bleed valve and the hydrogen concentration in an anode flow field;
- (6) FIG. 6 is an explanatory diagram showing a correspondence relationship between the power generation cell voltage and the opening and closing of the bleed valve;
- (7) FIG. 7 is a time chart for explaining an example of the operation in the first embodiment;
- (8) FIG. 8 is a flowchart for explaining the operation in the second embodiment; and
- (9) FIG. 9 is a time chart for explaining an example of the operation in the second embodiment.

DESCRIPTION OF THE INVENTION

- (10) Embodiments of a fuel cell system according to the present invention will be described below in detail with reference to the accompanying drawings.

First Embodiment

- (11) [Configuration]

- (12) FIG. 1 is a schematic block diagram showing the configuration of a fuel cell vehicle 10 in which a fuel cell system 14 according to a first embodiment is incorporated.
- (13) The fuel cell vehicle 10 includes a control device 12 for controlling the entire fuel cell vehicle 10, the fuel cell system 14, and an output unit 20 electrically connected to the fuel cell system 14.
- (14) The fuel cell system 14 is basically constituted by a fuel cell stack (also referred to simply as a fuel cell) 16, a hydrogen tank 18, an oxygen-containing gas-related device 22, and a fuel gas-related device 24.
- (15) The oxygen-containing gas-related device 22 includes an air pump 26 and a humidifier (HUM) 28.
- (16) The fuel gas-related device 24 includes an injector (INJ) 30, an ejector (EJT) 32, and a gas-liquid separator 34.
- (17) In the fuel cell stack 16, a plurality of power generation cells 40 are stacked. The power generation cells 40 each include a membrane electrode assembly 44, and separators 45 and 46 that sandwich the membrane electrode assembly 44.
- (18) The membrane electrode assembly 44 includes, for example, a solid polymer electrolyte membrane 41 which is a perfluorosulfonic acid thin membrane containing water, and a cathode 42 and an anode 43 that sandwich the solid polymer electrolyte membrane 41.
- (19) The cathode 42 and the anode 43 each include a gas diffusion layer (not shown) made of carbon paper or the like. The porous carbon particles having a platinum alloy supported on the surface thereof are uniformly applied to the surface of the gas diffusion layer to form an electrode catalyst layer (not shown). The electrode catalyst layers are formed on both surfaces of the solid polymer electrolyte membrane 41.
- (20) A cathode flow field (oxygen-containing gas flow field) 47 that allows an oxygen-containing gas inlet communication port 116 and an oxygen-containing gas outlet communication port 102 to communicate with each other is formed on the surface of the separator 45 that faces the membrane

electrode assembly **44**.

(21) An anode flow field (fuel gas flow field) **48** that allows a fuel gas inlet communication port **146** and a fuel gas outlet communication port **148** to communicate with each other is formed on the surface of the separator **46** that faces the membrane electrode assembly **44**.

(22) In the anode **43**, when the fuel gas is supplied, hydrogen ions are generated from hydrogen molecules by an electrode reaction by the catalyst, and the hydrogen ions pass through the solid polymer electrolyte membrane **41** and move to the cathode **42**, while electrons are released from the hydrogen molecules.

(23) The electrons released from the hydrogen molecules move from a negative electrode terminal **86** through an output regulator **80** and an external load such as a motor **84** to the cathode **42** via a positive electrode terminal **88**.

(24) At the cathode **42**, hydrogen ions and electrons react with oxygen contained in the supplied oxygen-containing gas by the action of the catalyst to produce water.

(25) A voltage sensor **92** for detecting a generated voltage V_{fc} is provided between the positive electrode terminal **88** and the negative electrode terminal **86**. Furthermore, a current sensor **93** for detecting a generated current I_{fc} is provided in the wiring between the positive electrode terminal **88** and the output regulator **80**.

(26) The air pump **26** is formed of a mechanical supercharger or the like driven by an inverter and a motor (not shown) to which power of an energy storage device **82** is supplied, and has a function of sucking and pressurizing atmospheric air (air), and supplying it to the humidifier **28**.

(27) The humidifier **28** includes a flow path **52** through which the oxygen-containing gas (dry air) flows, and a flow path **54** through which exhaust gas from the cathode flow field **47** of the fuel cell stack **16** flows through the oxygen-containing gas outlet communication port **102** of the fuel cell stack **16** and a pipeline **104**.

(28) The exhaust gas from the cathode flow field **47** of the fuel cell stack **16** is humid oxygen-containing exhaust gas when a bleed valve **158** is closed or humid oxygen-containing exhaust gas mixed with fuel exhaust gas when the bleed valve **158** is opened.

(29) The pipeline **104** is provided with a cathode outlet pressure sensor **69** that detects the pressure of the exhaust gas flowing through the pipeline **104** as a cathode outlet pressure P_{co} [kPa].

(30) The humidifier **28** has a function of humidifying the oxygen-containing gas supplied from the air pump **26**. That is, the humidifier **28** transfers moisture contained in the exhaust gas to the supply gas (oxygen-containing gas) through the porous membrane.

(31) A suction port side of the air pump **26** communicates with the atmosphere through a pipeline **106**, an air flow sensor (AFS) (flow rate sensor) **56**, and a pipeline **108**.

(32) The air flow sensor **56** measures a mass flow rate M [g/min] of the oxygen-containing gas supplied from the air pump **26** to the cathode flow field **47** of the fuel cell stack **16**, and outputs the measured mass flow rate to the control device **12**.

(33) A discharge port side of the air pump **26** communicates with one end side of the flow path **52** of the humidifier **28** through pipelines **110** and **112**. The other end side of the flow path **52** of the humidifier **28** communicates with one end side of a pipeline **114** (oxygen-containing gas supply flow path), and the other end side of the pipeline **114** communicates with the cathode flow field **47** in the fuel cell stack **16** through the oxygen-containing gas inlet communication port **116**.

(34) The pipeline **110** is provided with a discharge pressure sensor **64** that detects the pressure of the oxygen-containing gas discharged from the air pump **26** as an oxygen-containing gas discharge pressure P_o [kPa].

(35) The pipeline **114** is provided with a cathode inlet pressure sensor **67** that detects the pressure of the oxygen-containing gas flowing through the pipeline **114** as a cathode inlet pressure P_{ci} [kPa].

(36) A discharge side of the flow path **54** of the humidifier **28** communicates with one inlet side of a diluter **66** through pipelines **117** and **118**.

(37) The pipeline **110** on the discharge port side of the air pump **26** branches such that one of the branched pipelines communicates with the pipeline **112**, and the other of the branched pipelines communicates with the pipeline **118** via a bypass pipeline **120** and a bypass valve **122**.

(38) The pipeline **118** is provided with a diluter inlet pressure sensor **62** that detects the pressure of gas flowing through the pipeline **118** as a diluter inlet pressure P_d [kPa].

(39) The hydrogen tank **18** includes an electromagnetically operated shut-off valve, and is a container that stores high-purity hydrogen compressed at high pressure.

(40) The fuel gas discharged from the hydrogen tank **18** is supplied to the inlet of the anode flow field **48** of the fuel cell stack **16** through a pipeline **140**, the injector **30**, a pipeline **142**, the ejector **32**, and a pipeline **144** (fuel gas supply flow path), via the fuel gas inlet communication port **146**.

(41) The pipeline **144** is provided with an anode inlet pressure sensor **68** that detects the pressure of the fuel gas flowing through the pipeline **144** as an anode inlet pressure P_{ai} [kPa].

(42) The outlet of the anode flow field **48** communicates with an inlet **151** of the gas-liquid separator **34** through the fuel gas outlet communication port **148** and a pipeline **150** (fuel exhaust gas flow path), and fuel exhaust gas (anode off-gas), which is hydrogen-containing gas, is supplied from the anode flow field **48** to the gas-liquid separator **34**.

(43) The pipeline **150** is provided with an anode outlet pressure sensor **70** that detects the pressure of the fuel exhaust gas flowing through the pipeline **150** as an anode outlet pressure P_{oa} [kPa], and an anode outlet temperature sensor **72** that detects the temperature of the fuel exhaust gas flowing through the pipeline **150** as an anode outlet temperature T_{oa} [$^{\circ}$ C.].

(44) The gas-liquid separator **34** separates the fuel exhaust gas into a gas component and a liquid component (liquid water). The gas component of the fuel exhaust gas is discharged from a gas discharge port **152** of the gas-liquid separator **34** and supplied to the ejector **32** through a pipeline **154**. On the other hand, when the bleed valve **158** is opened as necessary, the fuel exhaust gas is also supplied to the pipeline **114** for oxygen-containing gas through a communication pipeline **156** (communication flow path) and the bleed valve **158** (on-off valve).

(45) The bleed valve **158** is opened to prevent deterioration of the anode **43** caused by a decrease in the hydrogen concentration in the anode flow field **48** due to nitrogen gas present in the cathode flow field **47** permeating through the membrane electrode assembly **44**. That is, the bleed valve **158** is opened when it is determined that the hydrogen concentration in the anode flow field **48** has decreased during normal power generation such as during traveling.

(46) The fuel exhaust gas supplied to the pipeline **114** for oxygen-containing gas is mixed, in the pipeline **114**, with the oxygen-containing gas supplied from the air pump **26**, and is supplied to the cathode flow field **47** of the fuel cell stack **16** through the oxygen-containing gas inlet communication port **116**.

(47) A part of the fuel exhaust gas supplied to the cathode flow field **47** is hydrogen-ionized by the catalytic reaction of the cathode **42**, and the hydrogen ions react with the oxygen-containing gas to produce water. The remaining unreacted fuel exhaust gas is discharged from the oxygen-containing gas outlet communication port **102**, flows through the pipeline **104**, the flow path **54**, and the pipelines **117** and **118**, is diluted in the diluter **66** by the oxygen-containing gas supplied from the bypass pipeline **120**, and is discharged to the outside (atmosphere) of the fuel cell vehicle **10** through a pipeline **124**.

(48) Fuel gas is supplied to the ejector **32** from the injector **30** provided on the upstream side of the ejector **32**, through the pipeline **142**. Therefore, the fuel exhaust gas (the gas component) supplied via the gas-liquid separator **34** is sucked by the ejector **32** and mixed with the fuel gas, and, in this state, supplied to the anode flow field **48** of the fuel cell stack **16** via the fuel gas inlet communication port **146** through the pipeline **144** of the fuel cell stack **16**.

(49) The liquid component of the fuel exhaust gas is discharged from a liquid discharge port **160** of the gas-liquid separator **34** to the outside of the fuel cell vehicle **10** through a pipeline **162**, a drain valve **164**, a pipeline **166**, the diluter **66**, and the pipeline **124**.

(50) Actually, a part of the fuel exhaust gas is discharged from the drain valve **164** to the pipeline **166** together with the liquid component. In order to discharge the fuel exhaust gas to the outside after diluting the hydrogen gas in the fuel exhaust gas, a part of the oxygen-containing gas discharged from the air pump **26** is supplied to the diluter **66** through the bypass pipeline **120** and the pipeline **118**.

(51) Therefore, after the hydrogen gas in the fuel exhaust gas is diluted in the diluter **66**, the fuel exhaust gas is discharged to the outside.

(52) The fuel cell stack **16** is further provided with a coolant supply flow path **74a** and a coolant discharge flow path **74b** respectively for supplying and discharging a coolant to and from a coolant flow field (not shown) provided in the fuel cell stack **16**. A temperature sensor **76** is provided in the coolant discharge flow path **74b**, and the temperature sensor **76** measures a temperature T_w [$^{\circ}$ C.] of the coolant flowing through the coolant discharge flow path **74b** as the temperature of the fuel cell stack **16**.

(53) The control device **12** is constituted by an electronic control unit (ECU) including a microcomputer including a CPU (not shown), a storage unit (ROM and RAM), and the like.

(54) The storage unit stores, for example, control programs for the fuel cell vehicle **10** and the fuel cell system **14**.

(55) The CPU of the control device **12** performs calculation according to the control programs to control the operation of the fuel cell vehicle **10** and the fuel cell system **14**.

(56) In order to perform operation control, the control device **12** adjusts the power supply ratio of the generated power of the fuel cell stack **16** and the power of the energy storage device **82** through the output regulator **80**, based on an accelerator position θ_a from an accelerator position sensor that detects the position of an accelerator pedal (not shown), a vehicle speed V_s from a vehicle speed sensor (not shown), an SOC (remaining capacity) from an SOC sensor **83** that detects the SOC of the energy storage device **82**, and the like, and controls driving of the motor **84**.

(57) Further, when controlling the driving of the motor **84**, the control device **12** controls the power generation of the fuel cell system **14** by controlling driving of the components constituting the oxygen-containing gas-related device **22** and the fuel gas-related device **24** based on the supply ratio of the generated power and based on the detection signals received from various sensors in addition to the pressure sensors such as the discharge pressure sensor **64** and the temperature sensors such as the temperature sensor **76**.

(58) As shown in FIG. 2A, the control device **12** also operates as a current converter **90** that outputs a generated current command value I_{fcom} to the output regulator **80** based on a generated power command value P_{fcom} and a generated voltage V_{fc} of the fuel cell stack **16**.

(59) The control device **12** may not be a single device and may be divided into two or more units, such as an energy management unit that determines the generated power command value P_{fcom} of the fuel cell stack **16**, a power generation control unit that includes the current converter **90** and controls power generation of the fuel cell stack **16** based on the generated power command value P_{fcom} , and a motor control unit that controls driving of the motor **84**.

(60) [Operation]

(61) Next, the operation of the fuel cell system **14** basically configured as described above will be described in the following order: [1] the operation during normal power generation such as during travel of the fuel cell vehicle **10**, etc. (the same operation is performed in a second embodiment described later) [2] the operation of setting the generated current command value I_{fcom} according to the first embodiment in consideration of a decrease in the hydrogen concentration in the anode flow field **48**, and [3] the operation of setting the generated current command value I_{fcom} according to the second embodiment in consideration of a decrease in the hydrogen concentration in the anode flow field **48**.

(62) [1] Description of Operation During Normal Power Generation Such as During Travel of Fuel Cell Vehicle **10**, Etc.

(63) The operation during normal power generation (such as during traveling in the case where the hydrogen concentration on the anode **43** side is in a state that does not require the bleed valve **158** to be opened) will be described below with reference to FIG. **3** in which the flow of fuel gas, the flow of oxygen-containing gas, and the flow of power are indicated by solid line arrows. In FIG. **3**, broken line arrows indicate the flow of fuel exhaust gas during opening of the bleed valve **158** described later.

(64) During normal power generation, the bleed valve **158** is closed, and the bypass valve **122** is opened primarily for dilution when the drain valve **164** is opened.

(65) On the oxygen-containing gas-related device **22** side, oxygen-containing gas is discharged from the air pump **26** which operates under the control of the control device **12** by using the high voltage power of the energy storage device **82** as a power source. The oxygen-containing gas discharged from the air pump **26** is humidified through the pipelines **110** and **112** and the humidifier **28**, and then supplied through the pipeline **114** to the oxygen-containing gas inlet communication port **116** of the fuel cell stack **16**.

(66) The humidifier **28** transfers moisture contained in the humid exhaust gas to the supply gas (oxygen-containing gas) flowing through the flow path **52** through the porous membrane, the humid exhaust gas being discharged from the oxygen-containing gas outlet communication port **102** of the fuel cell stack **16** to the pipeline **104** and flowing through the flow path **54**.

(67) On the other hand, on the fuel gas-related device **24** side, fuel gas (hydrogen gas) is supplied from the high-pressure hydrogen tank **18** to the pipeline **142** under the valve opening action of the injector **30**. The fuel gas is mixed with the fuel exhaust gas sucked by the ejector **32** through the pipeline (circulation path) **154**, passes through the ejector **32**, and is supplied to the fuel gas inlet communication port **146** of the fuel cell stack **16**.

(68) In the fuel cell stack **16**, the oxygen-containing gas is supplied from the oxygen-containing gas inlet communication port **116** to the cathode **42** through the cathode flow field **47** of each power generation cell **40**. On the other hand, hydrogen gas is supplied from the fuel gas inlet communication port **146** to the anode **43** through the anode flow field **48** of each power generation cell **40**. Therefore, in each power generation cell **40**, the oxygen-containing gas contained in the air supplied to the cathode **42** and the hydrogen gas supplied to the anode **43** are partially consumed by an electrochemical reaction (fuel cell reaction) in the electrode catalyst layer to generate power.

(69) Then, the reaction product water and the cathode exhaust gas made of the air supplied to and consumed by the cathode **42** are discharged from the oxygen-containing gas outlet communication port **102**, and flow through the pipeline **104** and the flow path **54** to the pipeline **117**. The cathode exhaust gas and the produced water flowing through the pipeline **117** are combined with the oxygen-containing gas supplied from the bypass pipeline **120**, are supplied through the pipeline **118** to the diluter **66**, and are discharged through the diluter **66** to the outside of the fuel cell vehicle **10** via the pipeline **124**.

(70) The hydrogen gas supplied to and consumed by the anode **43** is discharged from the fuel gas outlet communication port **148** as fuel exhaust gas (partially consumed fuel gas).

(71) The fuel exhaust gas is introduced into the gas-liquid separator **34** through the pipeline **150** to remove the liquid component (liquid water) thereof, and is then sucked by the ejector **32** through the pipeline **154** to be used for power generation reaction in the fuel cell stack **16**.

(72) High-voltage power generated by the fuel cell stack **16** in which a plurality of power generation cells **40** are electrically connected in series is stored in the energy storage device **82** through the output regulator **80**.

(73) The energy management unit of the control device **12** determines a generated power command value P_{fcom} of the fuel cell stack **16** in accordance with load states such as the accelerator position θ_a of the fuel cell vehicle **10**, the vehicle speed V_s , the SOC of the energy storage device **82**, and the set temperature of the air conditioner (not shown).

(74) The current converter **90** in the power generation control unit of the control device **12**

calculates a generated current command value I_{fcom} based on a generated power command value P_{fcom} and a generated voltage V_{fc} . The current converter **90** controls the oxygen-containing gas-related device **22**, the fuel gas-related device **24**, and the output regulator **80** so that a generated current I_{fc} corresponding to the generated current command value I_{fcom} is generated.

(75) In this case, the fuel cell vehicle **10** travels when the motor **84** is driven by the power of the energy storage device **82** and/or the fuel cell stack **16** under the control of the output regulator **80**.

(76) [2] Description of Operation of Setting Generated Current Command Value I_{fcom} According to First Embodiment in Consideration of Decrease in Hydrogen Concentration in Anode Flow Field **48**

(77) This will be described with reference to the flowchart shown in FIG. **4**. Although it is (the CPU of) the control device **12** that executes the processing (control programs) according to the flowchart of FIG. **4**, it is troublesome to refer to the control device **12** each time, so it is referred to as necessary.

(78) In step **S1**, a generated power command value P_{fcom} to be generated in the fuel cell stack **16** is calculated based on the accelerator position θ_a , the vehicle speed V_s , and the SOC.

(79) In step **S2**, a generated voltage V_{fc} is acquired by the voltage sensor **92**.

(80) In step **S3**, the current converter **90** of the control device **12** calculates a generated current command value I_{fcom} based on the generated power command value P_{fcom} and the generated voltage V_{fc} with reference to expression (1).

$$I_{fcom} = P_{fcom} / V_{fc} \quad (1)$$

(81) In step **S4**, power generation control for the fuel cell stack **16** is performed so that a generated current I_{fc} detected by the current sensor **93** is equal to I_{fcom} , based on the generated current command value I_{fcom} .

(82) Next, in step **S5**, the concentration of the fuel gas (hydrogen) flowing through the anode flow field **48** is estimated.

(83) In this case, the cathode inlet pressure P_{ci} is acquired from the cathode inlet pressure sensor **67**, the anode inlet pressure P_{ai} is acquired from the anode inlet pressure sensor **68**, and the temperature T_w of the coolant is acquired from the temperature sensor **76**. The temperature T_w of the coolant is substantially equal to the (internal) temperature of the fuel cell stack **16**, that is, the temperature of the fuel gas (hydrogen) in the anode flow field **48**.

(84) Then, based on the temperature T_w of the coolant, the cathode inlet pressure P_{ci} , and the anode inlet pressure P_{ai} , the permeation amount of nitrogen from the cathode **42** side to the anode **43** side is estimated, whereby the concentration of the fuel gas (hydrogen) in the anode flow field **48** is estimated. The hydrogen concentration may be measured by providing a hydrogen concentration sensor in the pipeline **144** and/or the pipeline **150**.

(85) Next, based on the estimated hydrogen concentration, opening/closing control of the bleed valve **158** is performed in steps **S6** to **S9**.

(86) The opening/closing control of the bleed valve **158** will be described with reference to the time chart of FIG. **5** and the power generation schematic diagram of FIG. **6**. Note that, in the power generation schematic diagram of FIG. **6**, gas such as nitrogen except for oxygen and hydrogen is not shown.

(87) In the fuel cell stack **16**, when the concentration of the fuel gas in the anode **43** is lower than a predetermined concentration (a threshold B in FIG. **5**) (in other words, when the fuel gas is insufficient), the solid polymer electrolyte membrane **41** becomes deteriorated. Therefore, it is determined in step **S6** whether the hydrogen concentration is less than the threshold B .

(88) When the hydrogen concentration is less than the threshold B in the hydrogen concentration determination of step **S6** (step **S6**: YES), the bleed valve **158** is opened in step **S7**.

(89) When the bleed valve **158** is opened, the fuel exhaust gas discharged from the fuel gas outlet communication port **148** is supplied to the cathode flow field **47** from the oxygen-containing gas inlet communication port **116** through the pipeline **114** via the communication pipeline **156**.

(90) Therefore, during opening of the bleed valve **158**, as shown in “during opening of bleed valve” in FIG. **6**, the amount of the oxygen-containing gas decreases on the cathode flow field **47** side, the potential on the cathode **42** side decreases, and a generated cell voltage V_{cell} [V] decreases. As a result, the generated voltage V_{fc} of the fuel cell stack **16** decreases.

(91) In the first embodiment, even if the generated voltage V_{fc} decreases, the control is performed so that the generated current command value I_{fcom} increases under the action of the current converter **90** as described with reference to the expression (1). Therefore, it is possible to avoid or suppress a decrease in a generated power P_{fc} caused by the decrease in the generated voltage V_{fc} .

(92) Further, during opening of the bleed valve **158**, the fuel gas discharge amount of the injector **30** is increased to increase the amount of fuel gas supplied from the fuel gas inlet communication port **146** to the anode flow field **48** of the fuel cell stack **16**, thereby increasing the hydrogen concentration.

(93) Next, in step **S8**, in order to suppress a decrease in the power generation efficiency of the fuel cell system **14**, it is determined whether the hydrogen concentration in the anode flow field **48** has recovered to a threshold A (threshold A > threshold B) (see FIG. **5**).

(94) If the hydrogen concentration has not recovered to the threshold A (step **S8**: NO), the process returns to step **S1**, and the processes of steps **S1** to **S5**, step **S6** (in the case of YES), step **S7**, and step **S8** (in the case of NO) are repeated.

(95) When the hydrogen concentration has recovered to the threshold A (step **S8**: YES), the bleed valve **158** is closed in step **S9**.

(96) In this case, as shown in “during closing of bleed valve” in FIG. **6**, power generation of the power generation cell **40** returns to normal, and the generated cell voltage V_{cell} rises to a normal voltage and is thus recovered.

(97) When the generated voltage V_{fc} is recovered, the generated current command value I_{fcom} is controlled to decrease by the current converter **90** as described with reference to the expression (1). Therefore, it is possible to avoid a change in the generated power P_{fc} or suppress a decrease in the generated power P_{fc} caused by an increase of the generated voltage V_{fc} due to the recovery.

(98) FIG. **7** is a time chart showing an example of the operation in the first embodiment described with reference to the flowchart of FIG. **4**.

(99) According to the time chart of FIG. **7**, the generated power command value P_{fcom} is maintained at a constant value regardless of the passage of time.

(100) As shown in FIG. **7**, although the generated voltage V_{fc} decreases when the bleed valve **158** is opened, the control is performed in the order of step **S7**, step **S8** (in the case of NO), steps **S1** to **S6** (in the case of YES, see also FIG. **5**), and step **S7** in the flowchart of FIG. **4**. Therefore, in step **S4**, the generated current command value I_{fcom} is controlled to increase by an amount corresponding to the decrease, based on the expression (1) above. Therefore, the actual generated power P_{fc} of the fuel cell stack **16** does not fluctuate.

(101) As described above, according to the first embodiment, the generated current command value I_{fcom} is controlled by the current converter **90** so as to be a value obtained by dividing the generated power command value P_{fcom} by the generated voltage V_{fc} . Therefore, even if the bleed valve **158** is opened, there is no need to change the power generation control during closing of the bleed valve **158** (during normal control), and the control can be simplified.

(102) In the first embodiment, during opening of the bleed valve **158**, a portion of the fuel gas (including nitrogen) in the fuel exhaust gas that has not been consumed by the catalytic reaction on the cathode **42** side is discharged from the oxygen-containing gas outlet communication port **102** as indicated by broken line arrows in FIG. **3**.

(103) A portion of the fuel gas that has not been consumed and is discharged from the oxygen-containing gas outlet communication port **102** is mixed with the oxygen-containing gas by opening the bypass valve **122**, and diluted and discharged to the atmosphere through the diluter **66** and the pipeline **124**.

(104) In this case, it is necessary to discharge, from the air pump **26**, extra amount of the oxygen-containing gas corresponding to the amount of the oxygen-containing gas flowing through the bypass valve **122**. However, since the amount of the fuel gas that needs to be diluted is reduced due to the electrode reaction at the cathode **42**, it is possible to suppress an increase in the discharge amount of the oxygen-containing gas from the air pump **26**.

(105) As a result, it is possible to improve the merchantability related to noise and vibration of the fuel cell vehicle **10** caused by noise and vibration of the air pump **26**.

(106) [3] Description of Operation of Setting Generated Current Command Value Ifccom According to Second Embodiment in Consideration of Decrease in Hydrogen Concentration in Anode Flow Field **48**

(107) Referring to the flowchart of FIG. **8**, the operation of setting the generated current command value Ifccom according to the second embodiment will be described. In the flowchart of FIG. **8**, the same step number is assigned to a process (step) corresponding to the process (step) shown in the flowchart of FIG. **4**, and a detailed description thereof is omitted.

(108) In the second embodiment, as shown in the schematic block diagram of FIG. **2B**, the current converter **90** in FIG. **1** in the first embodiment is replaced by a current converter **90A** including the current converter **90** described with reference to FIG. **2A** and a power/current conversion map **91**.

(109) In step **S1**, as in the first embodiment, a generated power command value Pfccom to be generated in the fuel cell stack **16** is calculated based on the accelerator position θ_a , the vehicle speed V_s , and the SOC.

(110) Next, in step **Sp**, when the bleed valve **158** is closed, as shown in FIG. **2B**, first power generation control is performed in which a generated current command value Ifccom is set based on the generated power command value Pfccom by using the power/current conversion map **91** (characteristics of the generated current relative to the generated power of the fuel cell stack **16**). In this case, power generation control for the fuel cell stack **16** is performed so that the generated current Ifc detected by the current sensor **93** is equal to Ifccom.

(111) Next, in step **S5**, the hydrogen concentration in the anode flow field **48** is estimated in the same manner as in the first embodiment.

(112) Next, when the hydrogen concentration is less than the threshold B in the hydrogen concentration determination in step **S6** (step **S6**: YES), prior to processing of opening the bleed valve **158** in step **S7**, the control is switched to second power generation control in which the generated current command value Ifccom is set by the current converter **90** based on the generated power command value Pfccom and the generated voltage Vfc in step **Sq** (see FIG. **2B**).

(113) Specifically, in the second power generation control in step **Sq**, a generated voltage Vfc is acquired by the voltage sensor **92** in the same manner as in step **S2** in the first embodiment. Next, in the same manner as in step **S3**, the current converter **90** in the current converter **90A** of the control device **12** calculates a generated current command value Ifccom based on the generated power command value Pfccom and the generated voltage Vfc with reference to the above expression (1). Further, in the same manner as in step **S4**, power generation control for the fuel cell stack **16** is performed so that the generated current Ifc detected by the current sensor **93** is equal to Ifccom based on the generated current command value Ifccom.

(114) Thus, during the opening of the bleed valve **158**, the control is continued in the order of step **S7**, step **S8** (in the case of NO, see also FIG. **5**), and step **Sq**, and the second power generation control is executed.

(115) FIG. **9** is a time chart showing an example of the operation in the second embodiment described with reference to the flowchart of FIG. **8**.

(116) According to the time chart of FIG. **9**, the generated power command value Pfccom is maintained at a constant value regardless of the passage of time, as in the first embodiment.

(117) As shown in FIG. **9**, when the bleed valve **158** is closed, in other words, when the generated voltage Vfc does not decrease due to shortage of the oxygen-containing gas, the processes of steps

S1 to S5 and step S6 (in the case of NO) are repeated after the process of step S9. Therefore, the actual generated power P_{fc} is maintained at the generated power command value P_{fcom} by the first power generation control that uses the power/current conversion map **91**.

(118) Further, when the bleed valve **158** is opened, the generated voltage V_{fc} decreases, but in this case, the processes of step S8 (in the case of NO), step Sq, and step S7 are executed and the current command value conversion logic is switched to the second power generation control. Therefore, the generated current command value I_{fcom} is increased by the current converter **90** in the current converter **90A** based on the above expression (1), so that the actual generated power P_{fc} is maintained at the generated power command value P_{fcom} .

(119) In the second embodiment, when the hydrogen concentration in the anode flow field **48** (on the anode **43** side) approaches the threshold B, the control is switched (step Sq) before the bleed valve **158** is actually opened (step S7) in consideration of the time lag of the bleed valve **158** which is an electromagnetic valve.

(120) It is when the fuel exhaust gas enters the cathode flow field **47** (cathode **42**) that the generated cell voltage V_{cell} decreases due to the opening of the bleed valve **158**. Therefore, there is basically no problem as long as the control is switched to the second power generation control after a little time has passed (for example, about 100 [ms]) from the opening of the bleed valve **158**.

(121) Similarly, the timing of returning to the first power generation control may be a timing at which the fuel exhaust gas does not enter the cathode flow field **47** (cathode **42**). In practice, the amount of the entering fuel exhaust gas becomes very small in 10 to 100 [ms].

(122) As described above, the slope of the increase or decrease of the hydrogen concentration in the anode flow field **48** (anode **43**) (see FIG. 5) can be estimated based on the internal temperature of the fuel cell stack **16** (the above-described temperature T_w of the coolant can be used instead), the cathode inlet pressure P_{ci} , and the anode inlet pressure P_{ai} .

(123) Therefore, since the timing of opening the bleed valve **158** is also predicted in consideration of this slope, the threshold B can be made slightly higher to cope with the situation.

(124) Also in the second embodiment, during opening of the bleed valve **158**, it is possible to suppress an increase in the discharge amount of the oxygen-containing gas from the air pump **26** for dilution, and to improve the merchantability related to noise and vibration of the fuel cell vehicle **10**, as described in the first embodiment.

INVENTIONS THAT CAN BE GRASPED FROM EMBODIMENTS

(125) The invention that can be grasped from the above embodiments will be described below. For convenience of understanding, some of the components are denoted by the reference numerals used in the above embodiments, but the components are not limited to those denoted by the reference numerals.

(126) The fuel cell system **14** according to the present invention comprises a fuel cell configured to generate power using a fuel gas supplied to an anode **43** through a fuel gas supply flow path and an oxygen-containing gas supplied to a cathode **42** through an oxygen-containing gas supply flow path, the fuel cell system **14** further comprising: a fuel exhaust gas flow path through which a fuel exhaust gas discharged from the anode flows; a connection flow path configured to allow the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other; an on-off valve configured to open or close the connection flow path; a control device **12** configured to control a state of supply of the fuel gas to the fuel cell and a state of supply of the oxygen-containing gas to the fuel cell; and a voltage acquirer configured to acquire a generated voltage (V_{fc}) of the fuel cell, wherein, when controlling a power generation state of the fuel cell based on a generated current command value I_{fcom} , the control device sets the generated current command value I_{fcom} based on a generated power command value P_{fcom} and the generated voltage V_{fc} acquired by the voltage acquirer.

(127) As a result, even when the on-off valve is opened to cause the fuel exhaust gas to flow through the cathode and the generated voltage of the fuel cell therefore decreases, the generated

current command value is increased based on the decrease in the generated voltage. Therefore, the generated power does not change. Since the power generation control is not changed in this control, the control can be simplified.

(128) Further, the fuel cell system according to the present invention comprises a fuel cell configured to generate power using a fuel gas supplied to an anode through a fuel gas supply flow path and an oxygen-containing gas supplied to a cathode through an oxygen-containing gas supply flow path, the fuel cell system further comprising: a fuel exhaust gas flow path through which a fuel exhaust gas discharged from the anode flows; a connection flow path configured to allow the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other; an on-off valve configured to open or close the connection flow path; a control device configured to control a state of supply of the fuel gas to the fuel cell and a state of supply of the oxygen-containing gas to the fuel cell; and a voltage acquirer configured to acquire a generated voltage of the fuel cell, wherein, when controlling a power generation state of the fuel cell based on a generated current command value, the control device performs first power generation control in which the generated current command value is set based on a generated power command value during closing of the on-off valve, and performs second power generation control in which the generated current command value is set based on the generated power command value and the generated voltage during opening of the on-off valve.

(129) As a result, during power generation, when the concentration of the fuel gas flowing through the anode decreases due to the influence of nitrogen gas permeating from the cathode side to the anode, and the on-off valve of the connection flow path allowing the fuel exhaust gas flow path and the oxygen-containing gas supply flow path to communicate with each other is therefore opened, the generated current command value is set based on the generated power command value and the generated voltage. Therefore, the generated power of the fuel cell does not change or the change of the generated power of the fuel cell can be suppressed even if the fuel gas concentration control (fuel gas concentration decrease prevention control) at the anode is performed.

(130) Further, the fuel cell system may further comprise a fuel gas concentration acquiring unit configured to acquire a concentration of the fuel gas flowing through the anode, and the control device may open the on-off valve when the concentration of the fuel gas acquired by the fuel gas concentration acquiring unit becomes equal to or less than a predetermined value.

(131) As a result, deterioration of the electrolyte membrane due to a decrease in the concentration of the fuel gas flowing through the anode can be suppressed.

(132) The fuel cell system may further comprise a fuel gas concentration acquiring unit configured to acquire a concentration of the fuel gas flowing through the anode, and the control device may switch from the first power generation control to the second power generation control before opening the on-off valve when the concentration of the fuel gas acquired by the fuel gas concentration acquiring unit becomes equal to or less than a predetermined value.

(133) This makes it possible to achieve both suppression of deterioration of the electrolyte membrane and appropriate control for the generated power command value.

(134) It is to be understood that the present invention is not limited to the above-described embodiments, and various configurations can be adopted therein based on the description of this specification.

Claims

1. A fuel cell system comprising: a fuel cell configured to generate power using a fuel gas supplied to an anode through a fuel gas supply flow path and an oxygen-containing gas supplied to a cathode through an oxygen-containing gas supply flow path, the fuel cell system further comprising: a fuel exhaust gas flow path through which a fuel exhaust gas discharged from the anode flows; a connection flow path configured to allow the fuel exhaust gas flow path and the

oxygen-containing gas supply flow path to communicate with each other, the fuel exhaust gas flowing through the connection flow path; an on-off valve configured to open or close the connection flow path; a control device configured to control a state of supply of the fuel gas to the fuel cell and a state of supply of the oxygen-containing gas to the fuel cell, the control device comprising one or more processors that execute computer-executable instructions stored in a memory; and a voltage acquirer configured to acquire a generated voltage of the fuel cell, wherein when controlling a power generation state of the fuel cell based on a generated current command value, the one or more processors execute the computer-executable instructions to cause the control device to: close the on-off valve to perform first power generation control in which the generated current command value is set based on a generated power command value; and open the on-off valve to perform second power generation control in which the generated current command value is set based on the generated power command value and the generated voltage during opening of the on-off valve so that the fuel exhaust gas that has flowed through the connection flow path and the oxygen-containing gas supply flow path is supplied to the cathode together with the oxygen-containing gas and an amount of the fuel gas to be supplied to the anode through the fuel gas supply flow path is increased.

2. The fuel cell system according to claim 1, further comprising a fuel gas concentration acquiring unit configured to acquire a concentration of the fuel gas flowing through the anode, wherein the one or more processors cause the control device to open the on-off valve when the concentration of the fuel gas acquired by the fuel gas concentration acquiring unit becomes equal to or less than a predetermined value.

3. The fuel cell system according to claim 1, further comprising a fuel gas concentration acquiring unit configured to acquire a concentration of the fuel gas flowing through the anode, wherein the one or more processors cause the control device to switch from the first power generation control to the second power generation control before opening the on-off valve when the concentration of the fuel gas acquired by the fuel gas concentration acquiring unit becomes equal to or less than a predetermined value.
